Multistage Residual Model Evaluation Report

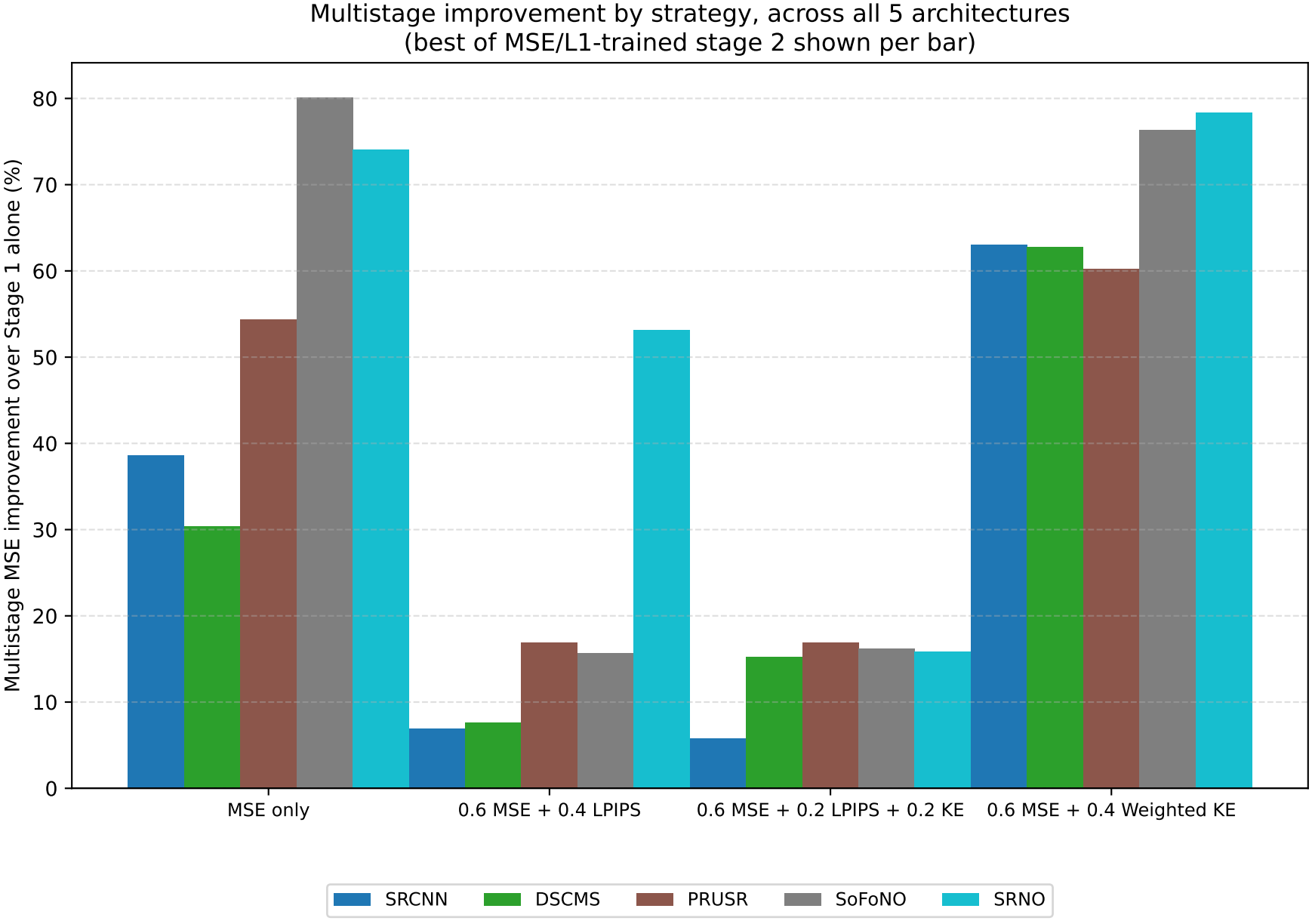
2026-09-23

Multistage residual model: final(LR) = stage1(LR) + epsilon * stage2(LR). Stage 1 is a frozen, already-trained model (any of the 4 loss strategies). Stage 2 is a freshly-initialized copy of the same architecture, trained with plain MSE or L1 to predict stage 1's normalized residual (HR - stage1(LR)) / epsilon, where epsilon = RMS(residual) over the training set.

All figures below use 'Final MSE': the validation MSE tracked during training itself (same metric EarlyStopping/checkpoint-selection use), not a separate full-test-set sweep -- directly comparable across every stage-1 loss strategy since the MSE component is logged separately regardless of LOSS_MODE.

Caveat: early stopping (patience=15, min_delta=1e-4) can trigger mechanically at exactly epoch 16 when a strategy's Final MSE range is smaller than min_delta itself (true for mse_lpips/mse_lpips_ke stage-1 models) -- this looks like fast convergence but may just be a metric-scale artifact cutting stage 2 off early, understating its true benefit for those two strategies.

Caveat: for mse_lpips_ke/mse_wke, stage 1 already directly optimizes KE/WKE, so its baseline KE/WKE values are small -- stage 2 (plain MSE/L1 on the residual, never KE/WKE-aware) consistently pushes them back up 2-19x across every architecture, even while MSE/SSIM/PSNR improve. Multistage trades spectral accuracy for pixel accuracy whenever stage 1 already nailed the spectrum.


Multistage improvement by strategy, across all 5 architectures
(best of MSE/L1-trained stage 2 shown per bar)
Multistage MSE improvement over Stage 1 alone (%)
0
10
20
30
40
50
60
70
80
MSE only
0.6 MSE + 0.4 LPIPS
0.6 MSE + 0.2 LPIPS + 0.2 KE
0.6 MSE + 0.4 Weighted KE
SRCNN
DSCMS
PRUSR
SoFoNO
SRNO

SRCNN -- all completed multistage configs

Strategy	Loss Norm	Epsilon	Stage-1 Final MSE	+Stage-2 Final MSE	Stage-2 Epochs	Improvement
MSE only	L1	0.03255	0.001081	0.000664	32	+38.6%
MSE only	MSE	0.03255	0.001081	0.000666	36	+38.4%
0.6 MSE + 0.4 LPIPS	L1	0.02274	**0.00052**	0.000485	16	+6.7%
0.6 MSE + 0.4 LPIPS	MSE	0.02274	0.00052	**0.000484**	16	+6.9%
0.6 MSE + 0.2 LPIPS + 0.2 KE	L1	0.02431	0.000592	0.00056	16	+5.4%
0.6 MSE + 0.2 LPIPS + 0.2 KE	MSE	0.02431	0.000592	0.000558	16	+5.7%
0.6 MSE + 0.4 Weighted KE	L1	0.05371	0.002889	0.001146	57	+60.3%
0.6 MSE + 0.4 Weighted KE	MSE	0.05371	0.002889	0.001067	54	**+63.1%**

DSCMS -- all completed multistage configs

Strategy	Loss Norm	Epsilon	Stage-1 Final MSE	+Stage-2 Final MSE	Stage-2 Epochs	Improvement
MSE only	L1	0.03079	0.001005	0.000806	25	+19.8%
MSE only	MSE	0.03079	0.001005	0.0007	34	+30.3%
0.6 MSE + 0.4 LPIPS	L1	0.01732	**0.000301**	0.000279	16	+7.3%
0.6 MSE + 0.4 LPIPS	MSE	0.01732	0.000301	**0.000278**	16	+7.6%
0.6 MSE + 0.2 LPIPS + 0.2 KE	L1	0.02101	0.000486	0.000413	16	+15.0%
0.6 MSE + 0.2 LPIPS + 0.2 KE	MSE	0.02101	0.000486	0.000412	16	+15.2%
0.6 MSE + 0.4 Weighted KE	L1	0.05251	0.002752	0.001024	104	**+62.8%**
0.6 MSE + 0.4 Weighted KE	MSE	0.05251	0.002752	0.001613	46	+41.4%

PRUSR -- all completed multistage configs

Strategy	Loss Norm	Epsilon	Stage-1 Final MSE	+Stage-2 Final MSE	Stage-2 Epochs	Improvement
MSE only	L1	0.02731	0.000769	0.000351	44	+54.4%
MSE only	MSE	0.02731	0.000769	0.000366	40	+52.4%
0.6 MSE + 0.4 LPIPS	L1	0.01816	**0.000332**	0.00028	16	+15.7%
0.6 MSE + 0.4 LPIPS	MSE	0.01816	0.000332	**0.000276**	16	+16.9%
0.6 MSE + 0.2 LPIPS + 0.2 KE	L1	0.02048	0.000443	0.000372	16	+16.0%
0.6 MSE + 0.2 LPIPS + 0.2 KE	MSE	0.02048	0.000443	0.000368	16	+16.9%
0.6 MSE + 0.4 Weighted KE	L1	0.03882	0.001427	0.000567	54	**+60.3%**
0.6 MSE + 0.4 Weighted KE	MSE	0.03882	0.001427	0.000579	48	+59.4%

SoFoNO -- all completed multistage configs

Strategy	Loss Norm	Epsilon	Stage-1 Final MSE	+Stage-2 Final MSE	Stage-2 Epochs	Improvement
MSE only	L1	0.02445	0.000629	0.000143	28	+77.3%
MSE only	MSE	0.02445	0.000629	0.000125	28	**+80.1%**
0.6 MSE + 0.4 LPIPS	L1	0.01037	**0.000115**	0.000101	16	+12.2%
0.6 MSE + 0.4 LPIPS	MSE	0.01037	0.000115	**9.7e-05**	16	+15.7%
0.6 MSE + 0.2 LPIPS + 0.2 KE	L1	0.01168	0.000136	0.000116	16	+14.7%
0.6 MSE + 0.2 LPIPS + 0.2 KE	MSE	0.01168	0.000136	0.000114	16	+16.2%
0.6 MSE + 0.4 Weighted KE	L1	0.02675	0.000707	0.000191	37	+73.0%
0.6 MSE + 0.4 Weighted KE	MSE	0.02675	0.000707	0.000167	30	+76.4%

SRNO -- all completed multistage configs

Strategy	Loss Norm	Epsilon	Stage-1 Final MSE	+Stage-2 Final MSE	Stage-2 Epochs	Improvement
MSE only	L1	0.02375	0.000575	0.000149	32	+74.1%
MSE only	MSE	0.02375	0.000575	0.000172	30	+70.1%
0.6 MSE + 0.4 LPIPS	L1	0.01173	0.000224	0.00011	16	+50.9%
0.6 MSE + 0.4 LPIPS	MSE	0.01173	0.000224	**0.000105**	16	+53.1%
0.6 MSE + 0.2 LPIPS + 0.2 KE	L1	0.01195	**0.000145**	0.000127	16	+12.4%
0.6 MSE + 0.2 LPIPS + 0.2 KE	MSE	0.01195	0.000145	0.000122	16	+15.9%
0.6 MSE + 0.4 Weighted KE	L1	0.02953	0.000891	0.000193	39	**+78.3%**
0.6 MSE + 0.4 Weighted KE	MSE	0.02953	0.000891	0.000208	41	+76.7%

SRCNN -- full test-set metrics (2022-2024): multistage value (% improvement over stage 1 alone)

Strategy	Loss Norm	MSE	SSIM	PSNR	LPIPS	FSIM	EPI	TKE	KE	WKE
MSE only	L1	0.0007209 (+38.4%)	0.9721 (+1.9%)	40.25 (+5.6%)	0.1617 (+29.5%)	0.7311 (+9.4%)	**0.03145 (-4.9%)**	0.0143 (+38.1%)	0.1902 (+56.4%)	0.2463 (+53.1%)
MSE only	MSE	0.0007234 (+38.2%)	0.9694 (+1.6%)	40.23 (+5.5%)	0.1692 (+26.2%)	0.7213 (+7.9%)	0.03104 (-6.1%)	0.01487 (+35.6%)	0.149 (+65.8%)	0.1821 (+65.3%)
0.6 MSE + 0.4 LPIPS	L1	0.0005294 (+7.2%)	**0.9772 (+0.1%)**	41.61 (+0.9%)	**0.07303 (-1.0%)**	**0.7693 (+0.4%)**	0.02839 (-0.2%)	0.009452 (+1.3%)	0.04065 (-0.5%)	0.05133 (-1.1%)
0.6 MSE + 0.4 LPIPS	MSE	**0.0005282 (+7.4%)**	0.9771 (+0.1%)	**41.62 (+0.9%)**	0.07305 (-1.0%)	0.7691 (+0.4%)	0.0284 (-0.2%)	**0.009436 (+1.5%)**	0.04198 (-3.8%)	0.05306 (-4.5%)
0.6 MSE + 0.2 LPIPS + 0.2 KE	L1	0.0006104 (+6.1%)	0.9744 (+0.0%)	41.01 (+0.7%)	0.07902 (-0.4%)	0.7531 (+0.2%)	0.02698 (-0.2%)	0.01158 (+1.9%)	**0.01443 (-172.8%)**	**0.01969 (-172.9%)**
0.6 MSE + 0.2 LPIPS + 0.2 KE	MSE	0.000608 (+6.5%)	0.9743 (+0.0%)	41.03 (+0.8%)	0.07907 (-0.4%)	0.7529 (+0.2%)	0.027 (-0.1%)	0.01155 (+2.1%)	0.01727 (-226.6%)	0.02372 (-228.7%)
0.6 MSE + 0.4 Weighted KE	L1	0.001244 (+60.2%)	0.9556 (+2.2%)	37.82 (+10.9%)	0.1978 (+23.1%)	0.6715 (+11.7%)	0.03042 (+32.6%)	0.02327 (+53.2%)	0.2512 (-1775.1%)	0.3226 (-1943.1%)
0.6 MSE + 0.4 Weighted KE	MSE	0.001152 (+63.1%)	0.9547 (+2.1%)	38.17 (+11.9%)	0.1957 (+23.9%)	0.6724 (+11.8%)	0.03059 (+33.3%)	0.02225 (+55.3%)	0.1535 (-1045.5%)	0.1776 (-1024.9%)

DSCMS -- full test-set metrics (2022-2024): multistage value (% improvement over stage 1 alone)

Strategy	Loss Norm	MSE	SSIM	PSNR	LPIPS	FSIM	EPI	TKE	KE	WKE
MSE only	L1	0.00088 (+16.3%)	0.9648 (+0.8%)	39.39 (+2.1%)	0.1864 (+9.5%)	0.6947 (+2.3%)	0.03248 (+1.1%)	0.01794 (+14.2%)	0.3264 (+24.3%)	0.4282 (+25.1%)
MSE only	MSE	0.0007619 (+27.5%)	0.9682 (+1.2%)	40.06 (+3.8%)	0.1722 (+16.4%)	0.716 (+5.4%)	**0.03266 (+1.6%)**	0.01606 (+23.2%)	0.1808 (+58.0%)	0.2243 (+60.8%)
0.6 MSE + 0.4 LPIPS	L1	0.0003041 (+7.2%)	**0.988 (+0.0%)**	44.02 (+0.8%)	**0.03881 (-0.7%)**	**0.8451 (+0.1%)**	0.02779 (-0.1%)	0.004164 (+0.9%)	0.02385 (-6.1%)	0.03595 (-6.4%)
0.6 MSE + 0.4 LPIPS	MSE	**0.0003029 (+7.6%)**	0.988 (+0.0%)	**44.04 (+0.8%)**	0.03882 (-0.7%)	0.8451 (+0.1%)	0.0278 (-0.0%)	**0.004149 (+1.3%)**	0.02397 (-6.6%)	0.03607 (-6.8%)
0.6 MSE + 0.2 LPIPS + 0.2 KE	L1	0.0004496 (+7.9%)	0.9828 (+0.0%)	42.31 (+0.9%)	0.05548 (-0.0%)	0.8036 (+0.1%)	0.02721 (-0.2%)	0.007134 (+2.2%)	**0.006238 (-113.0%)**	**0.008795 (-116.4%)**
0.6 MSE + 0.2 LPIPS + 0.2 KE	MSE	0.0004488 (+8.1%)	0.9827 (+0.0%)	42.32 (+1.0%)	0.0555 (-0.1%)	0.8036 (+0.1%)	0.02721 (-0.2%)	0.007107 (+2.6%)	0.008147 (-178.2%)	0.01161 (-185.5%)
0.6 MSE + 0.4 Weighted KE	L1	0.001101 (+62.9%)	0.9584 (+3.7%)	38.46 (+12.5%)	0.1624 (+39.0%)	0.6834 (+14.7%)	0.03006 (+8.2%)	0.02052 (+55.9%)	0.1063 (-940.8%)	0.1393 (-1118.7%)
0.6 MSE + 0.4 Weighted KE	MSE	0.001693 (+42.9%)	0.9313 (+0.8%)	36.58 (+7.0%)	0.2297 (+13.8%)	0.6215 (+4.3%)	0.02876 (+3.5%)	0.03319 (+28.7%)	0.09022 (-783.5%)	0.1174 (-926.8%)

PRUSR -- full test-set metrics (2022-2024): multistage value (% improvement over stage 1 alone)

Strategy	Loss Norm	MSE	SSIM	PSNR	LPIPS	FSIM	EPI	TKE	KE	WKE
MSE only	L1	0.0003787 (+54.1%)	0.984 (+2.1%)	43.08 (+8.7%)	0.09182 (+50.6%)	0.8033 (+14.7%)	**0.02865 (-7.4%)**	0.006587 (+61.8%)	0.0676 (+67.4%)	0.09591 (+62.1%)
MSE only	MSE	0.0003964 (+52.0%)	0.9816 (+1.9%)	42.86 (+8.1%)	0.09953 (+46.4%)	0.7867 (+12.3%)	0.02789 (-9.8%)	0.007443 (+56.8%)	0.05471 (+73.6%)	0.07582 (+70.1%)
0.6 MSE + 0.4 LPIPS	L1	0.0003021 (+16.0%)	**0.9863 (+0.3%)**	44.04 (+1.9%)	0.04429 (-8.9%)	0.8301 (+0.1%)	0.02792 (-0.2%)	0.004562 (+1.3%)	0.02084 (-2.4%)	0.03031 (-4.0%)
0.6 MSE + 0.4 LPIPS	MSE	**0.0002988 (+17.0%)**	0.9863 (+0.3%)	**44.09 (+2.0%)**	**0.04424 (-8.8%)**	**0.8302 (+0.1%)**	0.0279 (-0.3%)	**0.00453 (+2.0%)**	0.02016 (+0.9%)	0.0291 (+0.2%)
0.6 MSE + 0.2 LPIPS + 0.2 KE	L1	0.0004023 (+12.0%)	0.9821 (+0.2%)	42.81 (+1.4%)	0.05837 (-9.0%)	0.7946 (+0.2%)	0.02715 (-0.3%)	0.006932 (+2.7%)	**0.01434 (-427.0%)**	**0.0205 (-443.7%)**
0.6 MSE + 0.2 LPIPS + 0.2 KE	MSE	0.0003977 (+13.0%)	0.9821 (+0.2%)	42.86 (+1.5%)	0.05831 (-8.9%)	0.7947 (+0.2%)	0.02716 (-0.3%)	0.006887 (+3.4%)	0.01536 (-464.4%)	0.02196 (-482.5%)
0.6 MSE + 0.4 Weighted KE	L1	0.0006133 (+64.9%)	0.9751 (+2.4%)	40.95 (+12.3%)	0.1218 (+41.5%)	0.7471 (+12.8%)	0.02859 (+7.8%)	0.01163 (+60.0%)	0.04817 (-530.3%)	0.067 (-595.8%)
0.6 MSE + 0.4 Weighted KE	MSE	0.0006275 (+64.1%)	0.9735 (+2.2%)	40.86 (+12.0%)	0.1219 (+41.5%)	0.7423 (+12.1%)	0.02823 (+6.4%)	0.01205 (+58.5%)	0.04827 (-531.6%)	0.06573 (-582.7%)

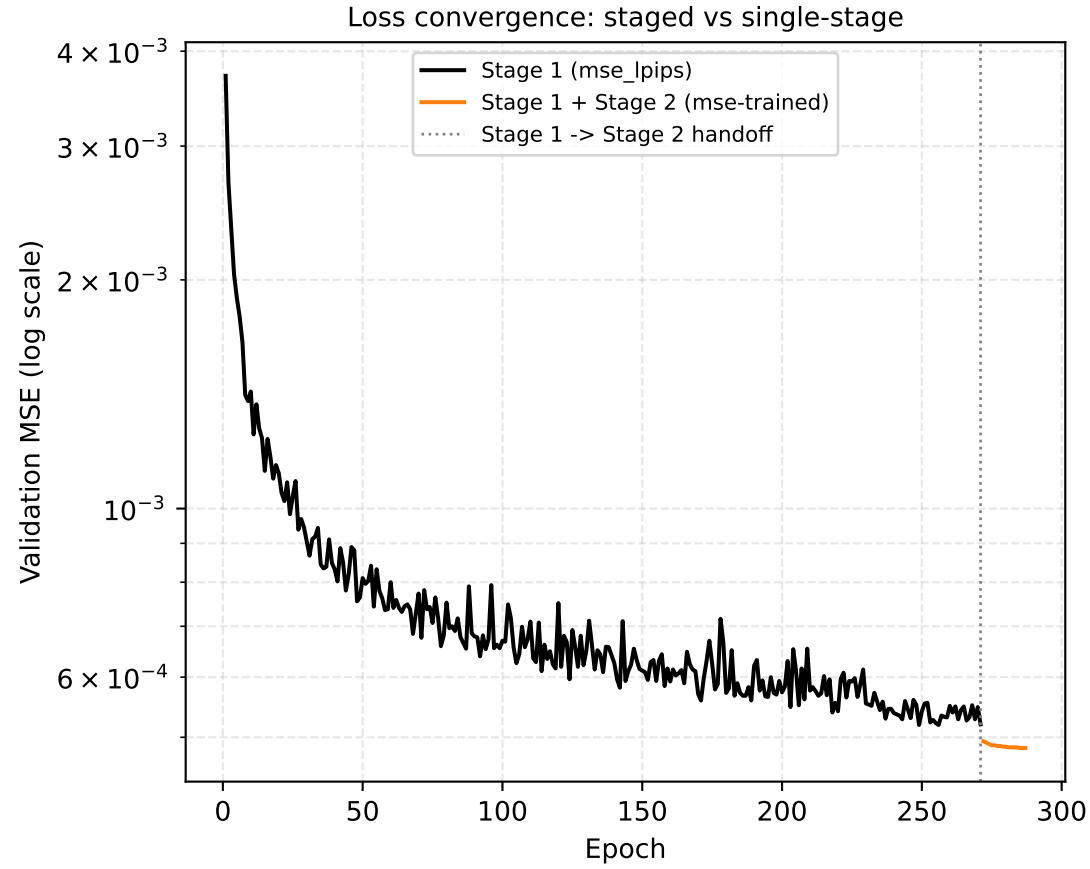
SoFoNO -- full test-set metrics (2022-2024): multistage value (% improvement over stage 1 alone)

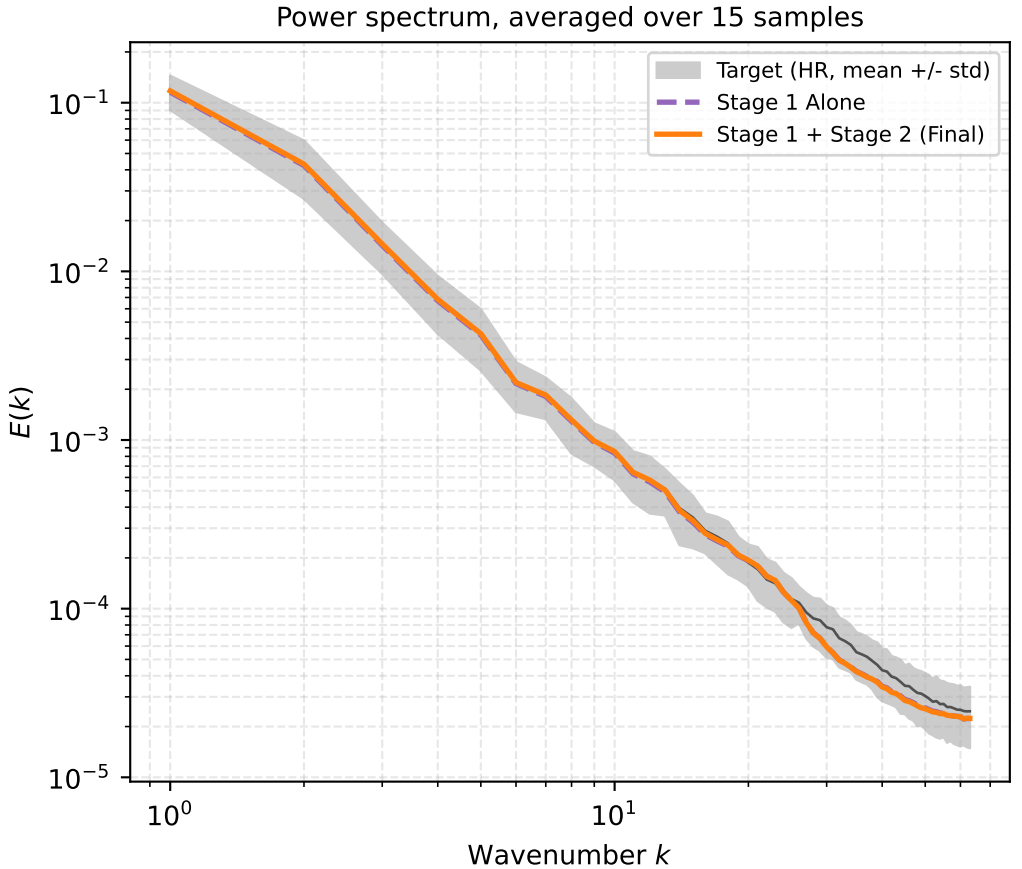
Strategy	Loss Norm	MSE	SSIM	PSNR	LPIPS	FSIM	EPI	TKE	KE	WKE
MSE only	L1	0.000156 (+76.5%)	0.9943 (+2.3%)	46.91 (+15.6%)	0.03093 (+80.7%)	0.9011 (+22.5%)	0.02831 (-5.3%)	0.001697 (+86.9%)	0.01642 (+87.6%)	0.02602 (+83.3%)
MSE only	MSE	0.0001363 (+79.5%)	0.9947 (+2.3%)	47.46 (+16.9%)	0.02665 (+83.4%)	0.9081 (+23.4%)	0.02827 (-5.5%)	0.001376 (+89.4%)	0.01486 (+88.8%)	0.02441 (+84.4%)
0.6 MSE + 0.4 LPIPS	L1	0.0001113 (+6.3%)	0.9955 (+0.0%)	48.36 (+0.6%)	**0.01318 (-0.7%)**	0.9259 (+0.2%)	0.02829 (-0.0%)	0.0008713 (+5.1%)	0.007246 (-6.0%)	0.01205 (-6.9%)
0.6 MSE + 0.4 LPIPS	MSE	**0.0001073 (+9.6%)**	**0.9955 (+0.0%)**	**48.52 (+1.0%)**	0.01321 (-0.9%)	**0.9261 (+0.2%)**	0.0283 (+0.0%)	**0.0008396 (+8.6%)**	0.0076 (-11.2%)	0.01271 (-12.9%)
0.6 MSE + 0.2 LPIPS + 0.2 KE	L1	0.0001276 (+13.0%)	0.995 (+0.0%)	47.79 (+1.3%)	0.01551 (+0.1%)	0.9175 (+0.2%)	**0.02834 (-0.2%)**	0.0011 (+9.9%)	**0.005232 (-376.9%)**	**0.008636 (-410.7%)**
0.6 MSE + 0.2 LPIPS + 0.2 KE	MSE	0.0001253 (+14.5%)	0.995 (+0.0%)	47.86 (+1.4%)	0.01552 (+0.0%)	0.9176 (+0.3%)	0.02832 (-0.2%)	0.001077 (+11.8%)	0.006233 (-468.1%)	0.01034 (-511.3%)
0.6 MSE + 0.4 Weighted KE	L1	0.0001967 (+75.0%)	0.9929 (+2.0%)	45.91 (+14.9%)	0.03783 (+70.5%)	0.8788 (+19.5%)	0.02817 (+0.8%)	0.002411 (+83.2%)	0.01516 (-430.5%)	0.02393 (-562.1%)
0.6 MSE + 0.4 Weighted KE	MSE	0.0001822 (+76.8%)	0.9926 (+2.0%)	46.22 (+15.7%)	0.03662 (+71.4%)	0.8774 (+19.4%)	0.02816 (+0.8%)	0.002285 (+84.1%)	0.01329 (-365.2%)	0.02104 (-481.9%)

SRNO -- full test-set metrics (2022-2024): multistage value (% improvement over stage 1 alone)

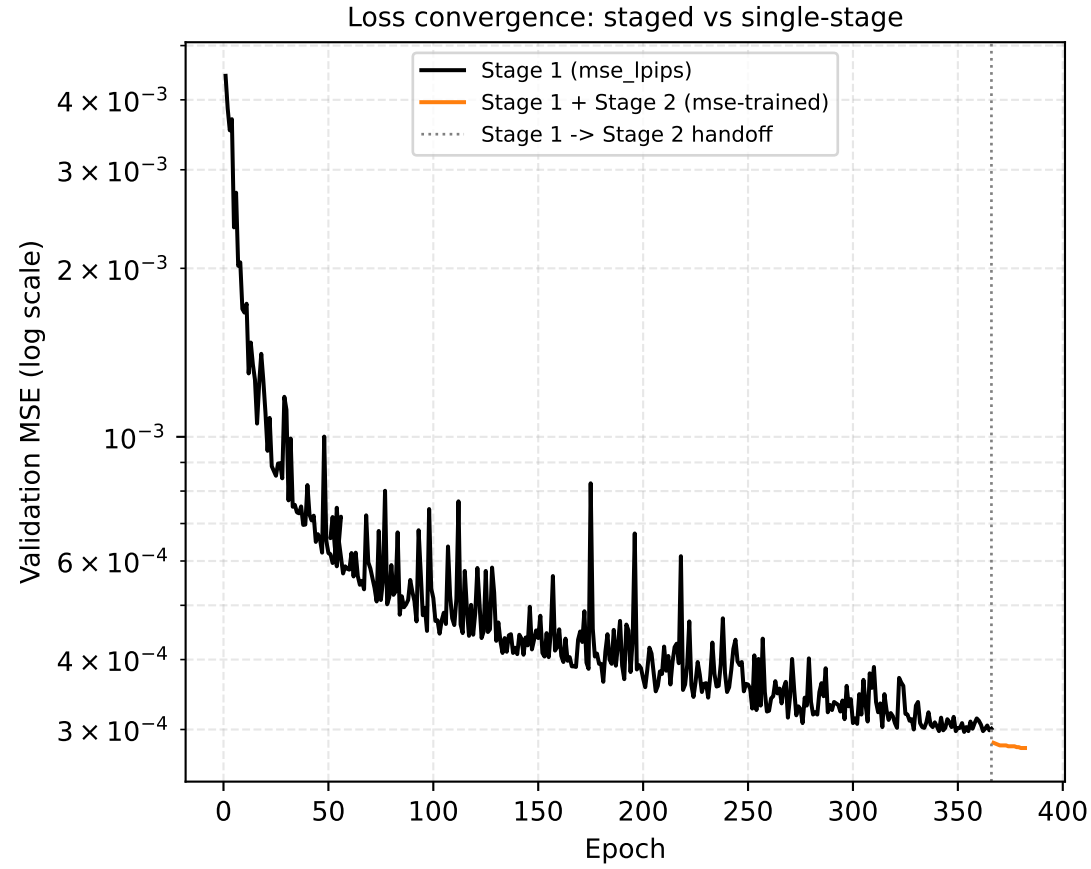
Strategy	Loss Norm	MSE	SSIM	PSNR	LPIPS	FSIM	EPI	TKE	KE	WKE
MSE only	L1	0.0001629 (+73.9%)	0.9941 (+2.2%)	46.73 (+14.4%)	0.03222 (+79.5%)	0.8996 (+21.9%)	**0.02835 (-4.6%)**	0.001819 (+85.6%)	0.02181 (+86.8%)	0.03491 (+83.3%)
MSE only	MSE	0.0001878 (+69.9%)	0.9922 (+2.0%)	46.08 (+12.8%)	0.04344 (+72.4%)	0.8771 (+18.8%)	0.02811 (-5.4%)	0.002426 (+80.9%)	0.02404 (+85.5%)	0.03768 (+81.9%)
0.6 MSE + 0.4 LPIPS	L1	0.000121 (+19.0%)	**0.9954 (+0.0%)**	48.01 (+2.1%)	0.01409 (-0.2%)	0.9223 (+0.3%)	0.02832 (+0.1%)	0.0009989 (+7.5%)	0.008117 (+11.4%)	0.01364 (+11.3%)
0.6 MSE + 0.4 LPIPS	MSE	**0.0001147 (+23.2%)**	0.9954 (+0.0%)	**48.24 (+2.6%)**	**0.01408 (-0.1%)**	**0.9224 (+0.3%)**	0.02833 (+0.1%)	**0.0009524 (+11.8%)**	0.008211 (+10.3%)	0.0138 (+10.3%)
0.6 MSE + 0.2 LPIPS + 0.2 KE	L1	0.0001406 (+10.5%)	0.9946 (+0.0%)	47.39 (+1.1%)	0.01666 (-0.3%)	0.9132 (+0.2%)	0.02824 (-0.2%)	0.001262 (+7.1%)	**0.005037 (-341.5%)**	**0.008288 (-363.6%)**
0.6 MSE + 0.2 LPIPS + 0.2 KE	MSE	0.0001345 (+14.4%)	0.9946 (+0.0%)	47.58 (+1.5%)	0.01664 (-0.2%)	0.9134 (+0.2%)	0.02826 (-0.1%)	0.00121 (+10.9%)	0.005143 (-350.8%)	0.008471 (-373.8%)
0.6 MSE + 0.4 Weighted KE	L1	0.0002105 (+76.7%)	0.9922 (+2.1%)	45.65 (+16.2%)	0.03983 (+70.8%)	0.8705 (+18.9%)	0.0283 (+3.9%)	0.002767 (+83.1%)	0.02187 (-583.7%)	0.03436 (-744.3%)
0.6 MSE + 0.4 Weighted KE	MSE	0.0002257 (+75.1%)	0.9908 (+1.9%)	45.3 (+15.3%)	0.04678 (+65.7%)	0.8566 (+17.1%)	0.0281 (+3.2%)	0.003163 (+80.6%)	0.02146 (-570.9%)	0.03342 (-721.4%)

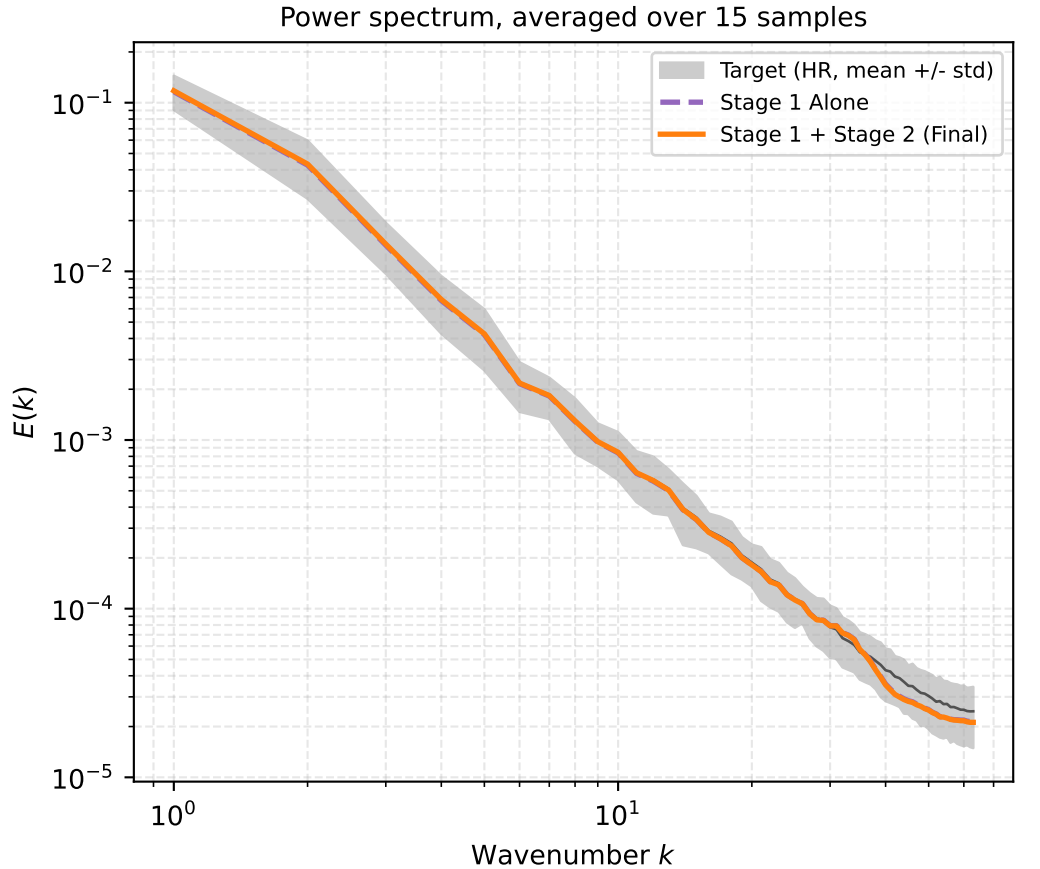
SRCNN -- best config: 0.6 MSE + 0.4 LPIPS (stage 2 MSE-trained), Final MSE 0.000484 (+6.9% vs stage 1 alone)



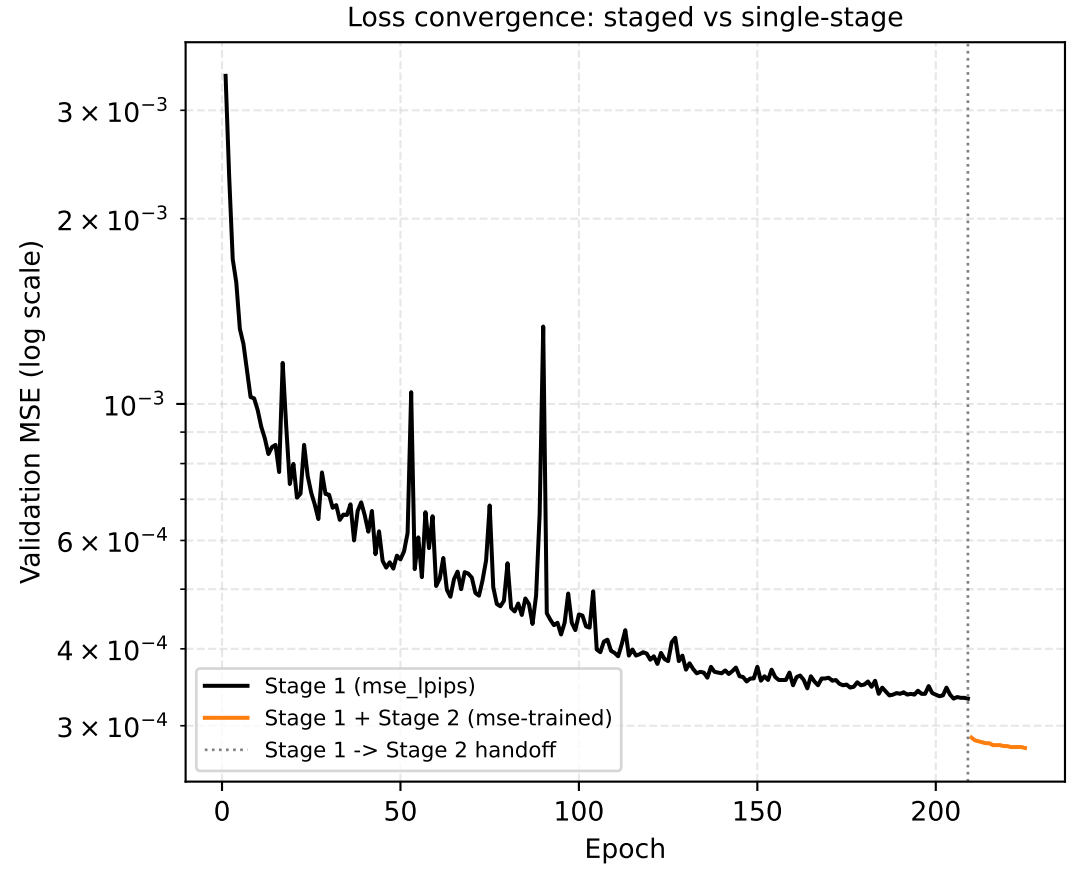


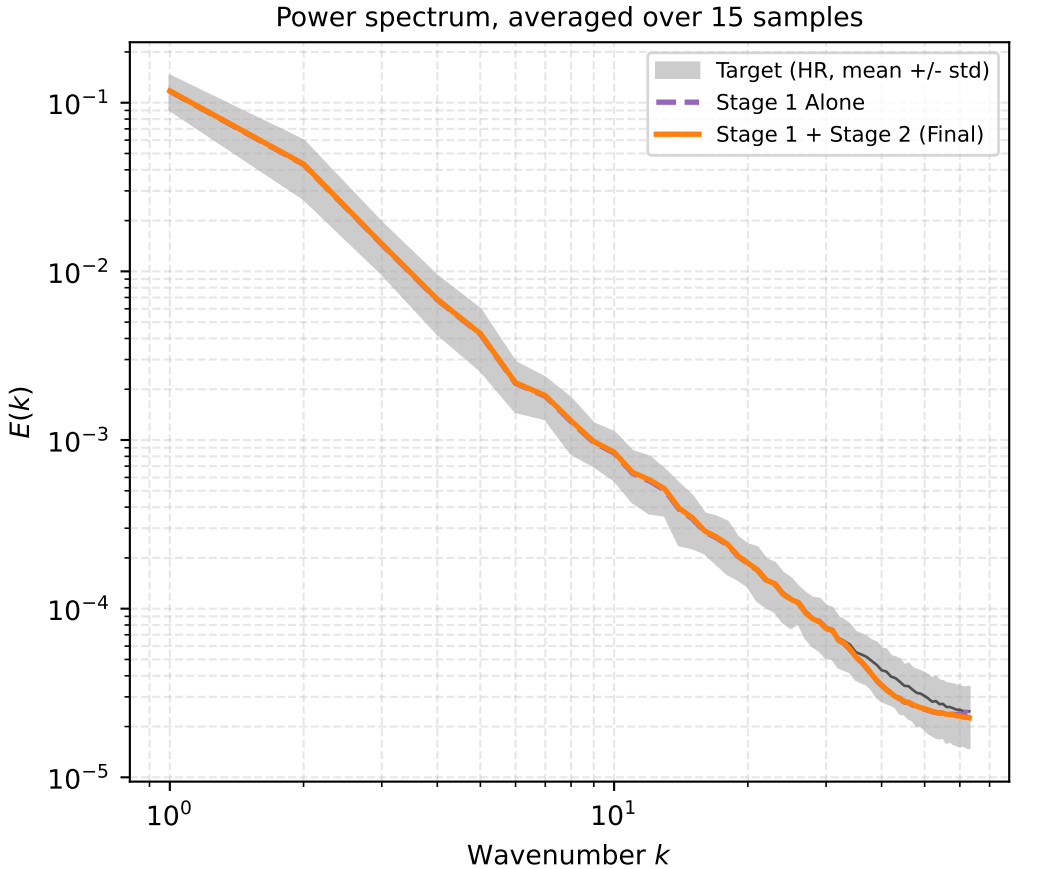
DSCMS -- best config: 0.6 MSE + 0.4 LPIPS (stage 2 MSE-trained), Final MSE 0.000278 (+7.6% vs stage 1 alone)

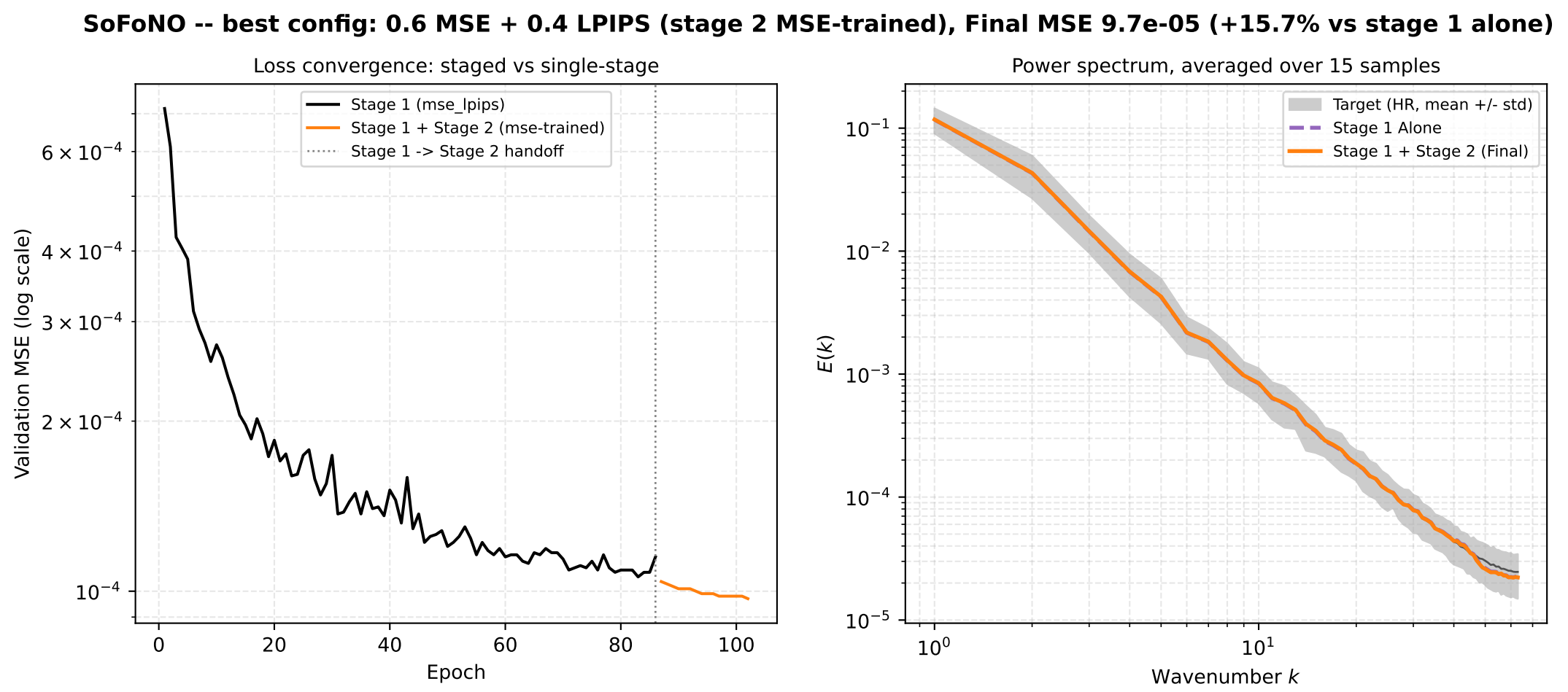




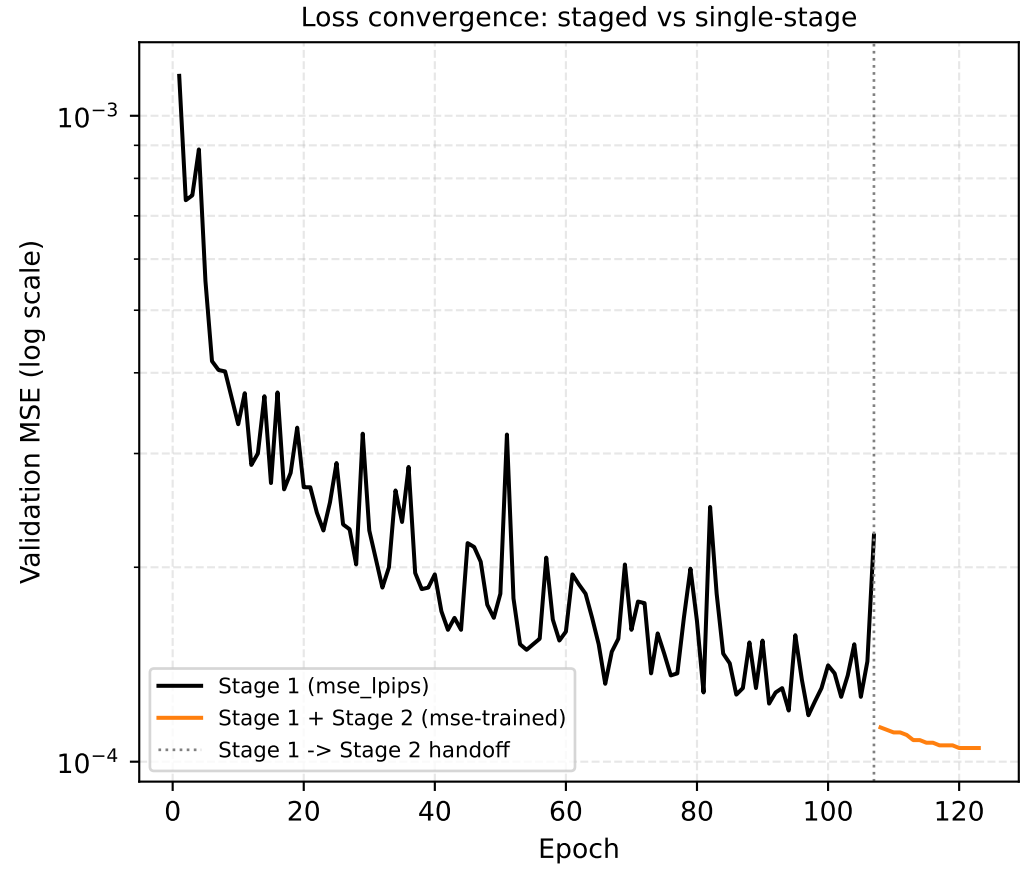
PRUSR -- best config: 0.6 MSE + 0.4 LPIPS (stage 2 MSE-trained), Final MSE 0.000276 (+16.9% vs stage 1 alone)

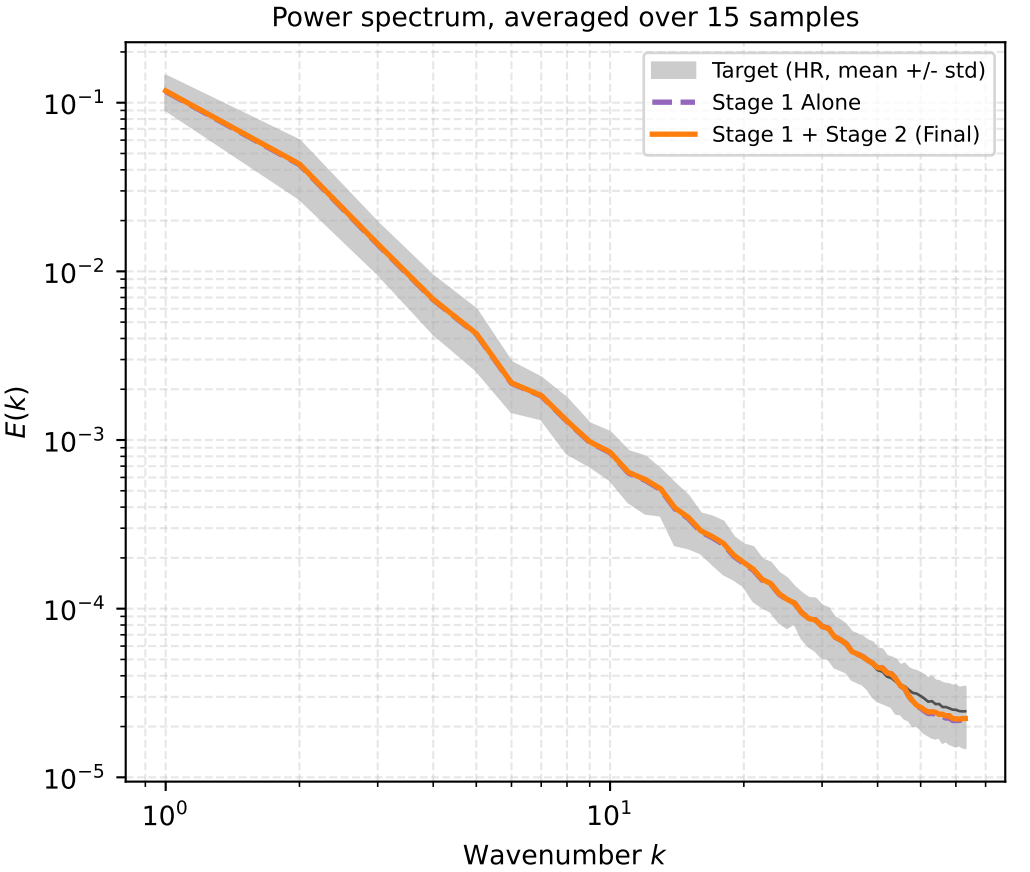


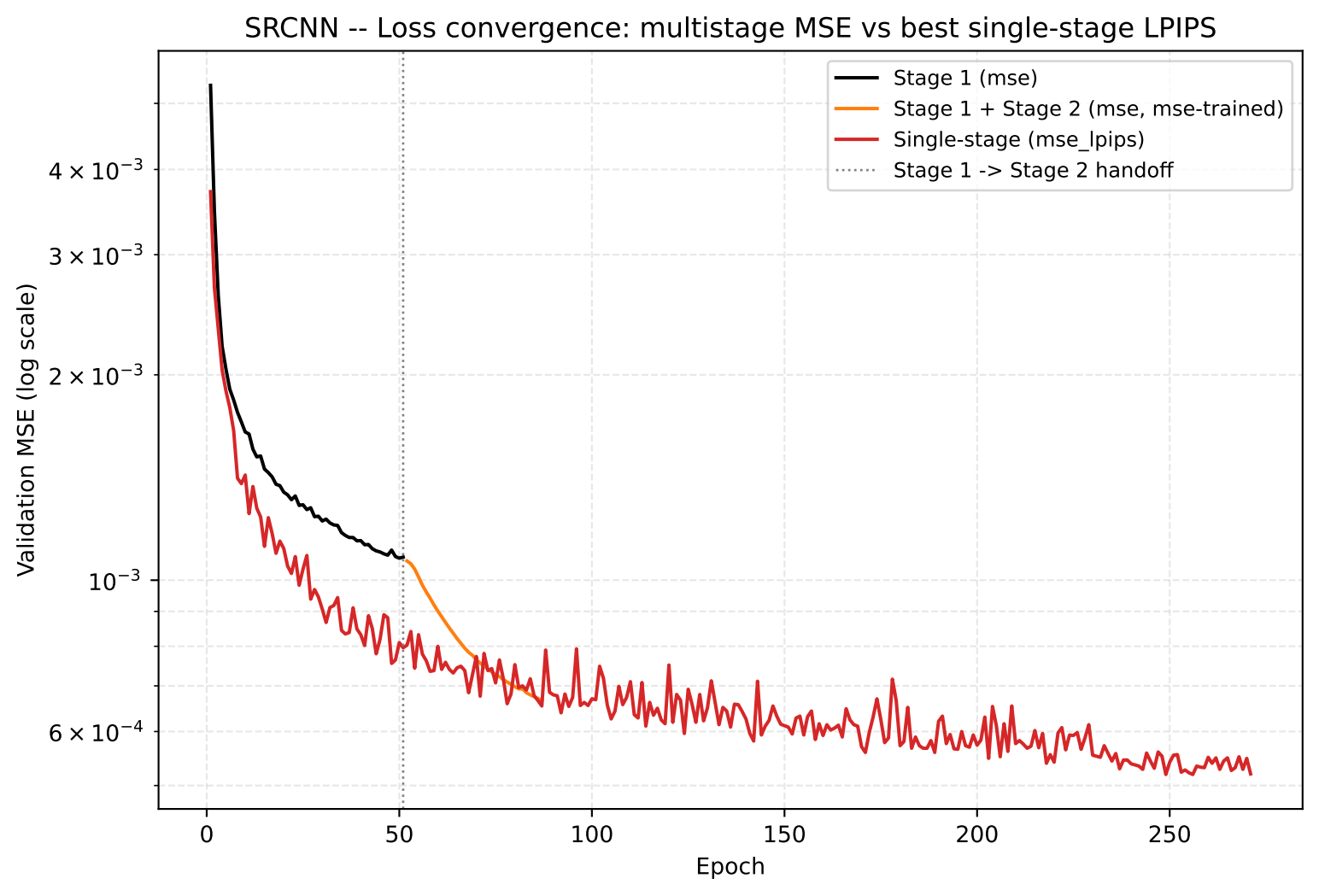


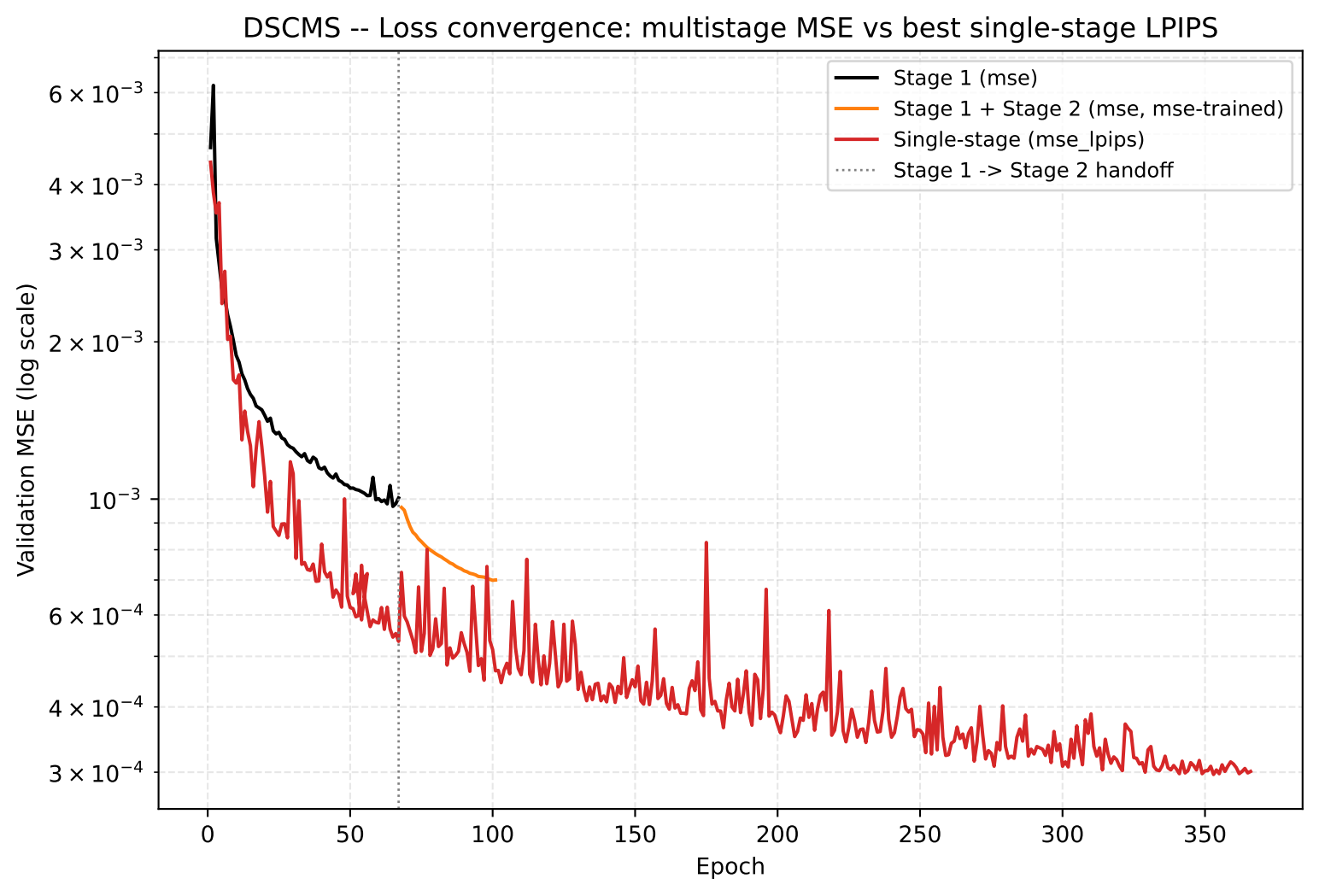

SoFoNO -- best config: 0.6 MSE + 0.4 LPIPS (stage 2 MSE-trained), Final MSE 9.7e-05 (+15.7% vs stage 1 alone)
Loss convergence: staged vs single-stage
Stage 1 (mse_lpips)
Stage 1 + Stage 2 (mse-trained)
Stage 1 -> Stage 2 handoff
Validation MSE (log scale)
Epoch
Power spectrum, averaged over 15 samples
Target (HR, mean +/- std)
Stage 1 Alone
Stage 1 + Stage 2 (Final)
E(k)
Wavenumber k

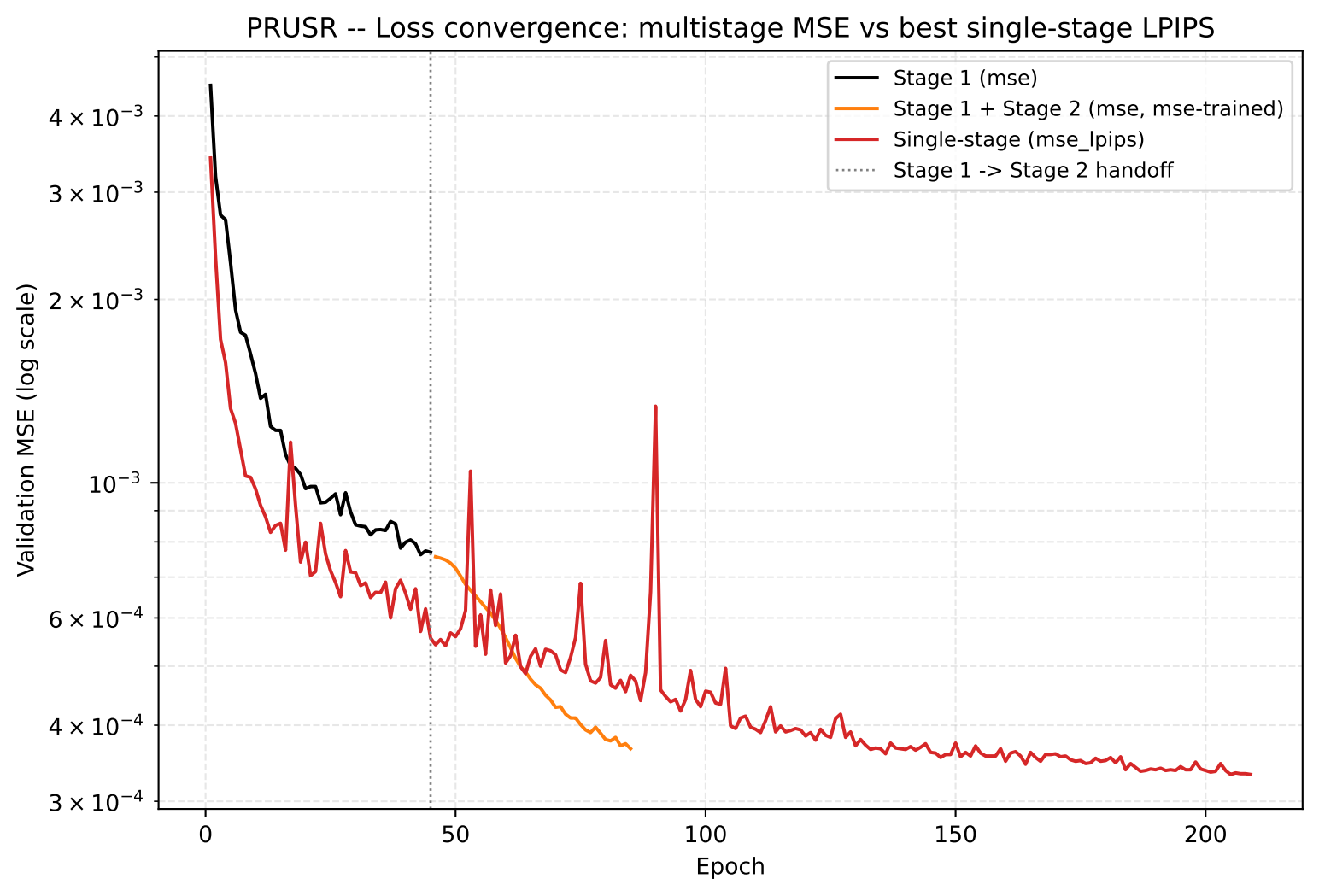
SRNO -- best config: 0.6 MSE + 0.4 LPIPS (stage 2 MSE-trained), Final MSE 0.000105 (+53.1% vs stage 1 alone)

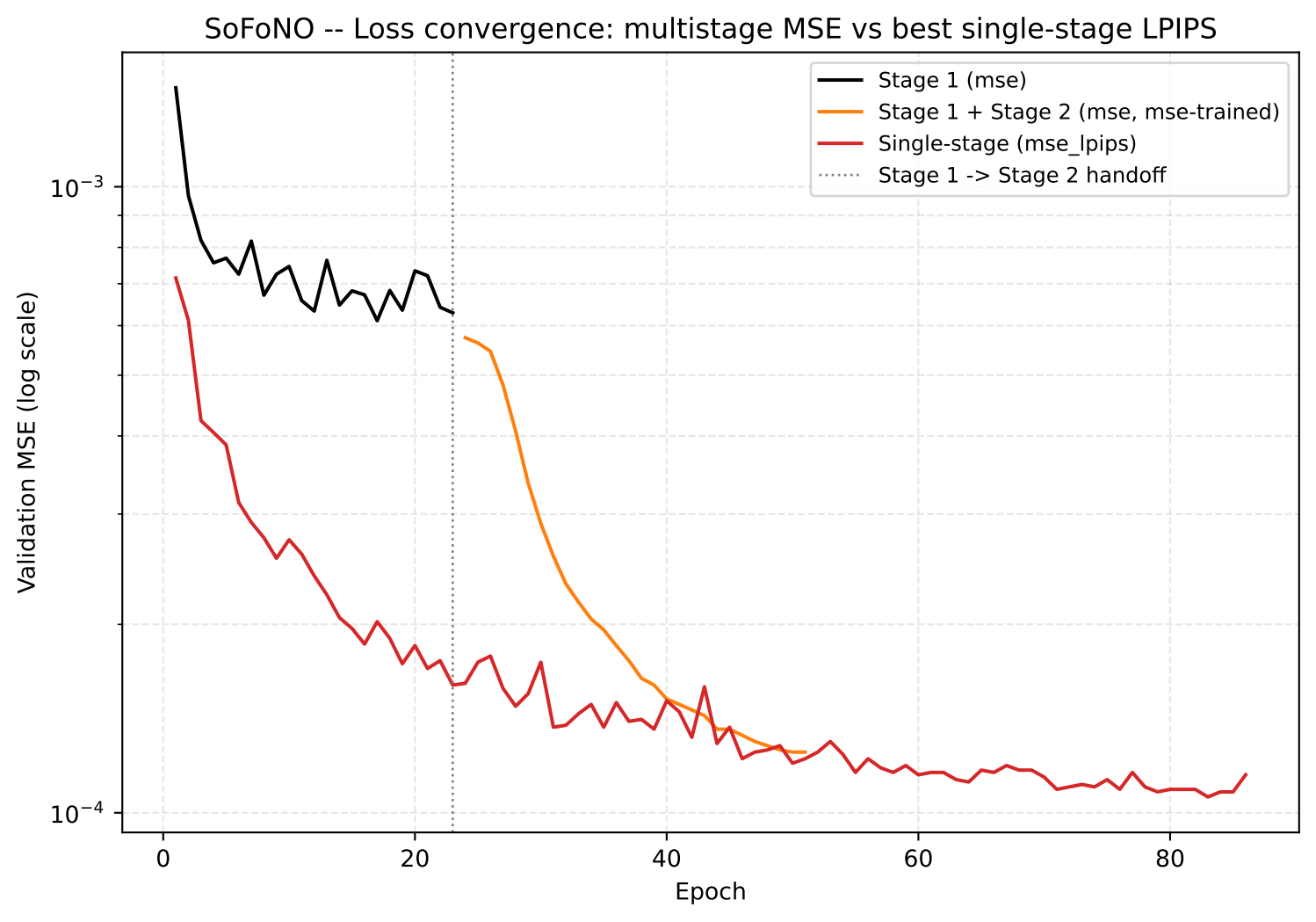


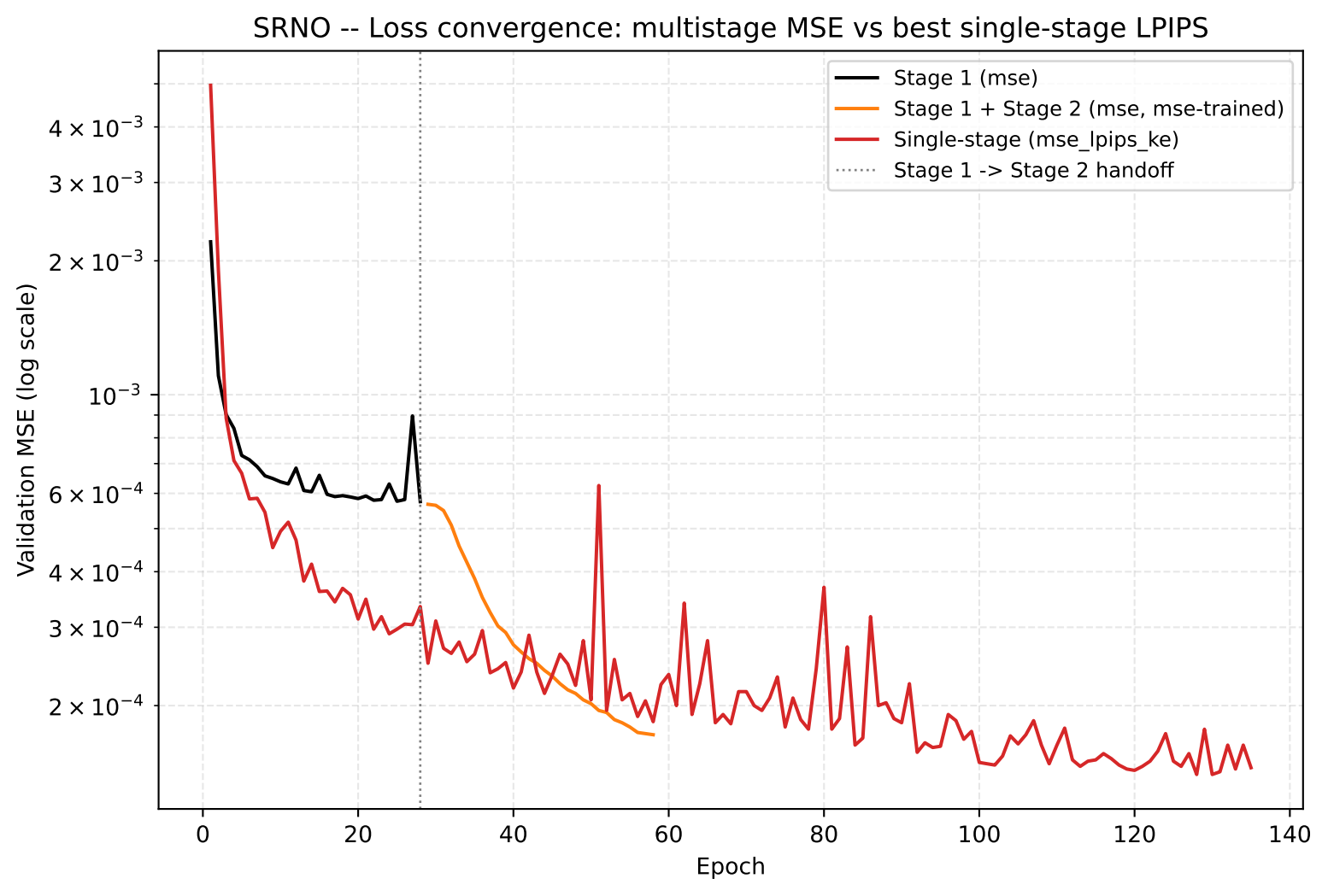



SRCNN -- Loss convergence: multistage MSE vs best single-stage LPIPS
Stage 1 (mse)
Stage 1 + Stage 2 (mse, mse-trained)
Single-stage (mse_lpips)
Stage 1 -> Stage 2 handoff
Validation MSE (log scale)
4×10^{-3}
3×10^{-3}
2×10^{-3}
10^{-3}
6×10^{-4}
0
50
100
150
200
250
Epoch


DSCMS -- Loss convergence: multistage MSE vs best single-stage LPIPS
Stage 1 (mse)
Stage 1 + Stage 2 (mse, mse-trained)
Single-stage (mse_lpips)
Stage 1 -> Stage 2 handoff
Validation MSE (log scale)
6×10^{-3}
4×10^{-3}
3×10^{-3}
2×10^{-3}
10^{-3}
6×10^{-4}
4×10^{-4}
3×10^{-4}
0
50
100
150
200
250
300
350
Epoch


PRUSR -- Loss convergence: multistage MSE vs best single-stage LPIPS
Stage 1 (mse)
Stage 1 + Stage 2 (mse, mse-trained)
Single-stage (mse_lpips)
Stage 1 -> Stage 2 handoff
Validation MSE (log scale)
4×10^{-3}
3×10^{-3}
2×10^{-3}
10^{-3}
6×10^{-4}
4×10^{-4}
3×10^{-4}
0
50
100
150
200
Epoch


SoFoNO -- Loss convergence: multistage MSE vs best single-stage LPIPS
Stage 1 (mse)
Stage 1 + Stage 2 (mse, mse-trained)
Single-stage (mse_lpips)
Stage 1 -> Stage 2 handoff
Validation MSE (log scale)
10^{-3}
10^{-4}
0
20
40
60
80
Epoch


SRNO -- Loss convergence: multistage MSE vs best single-stage LPIPS
Stage 1 (mse)
Stage 1 + Stage 2 (mse, mse-trained)
Single-stage (mse_lpips_ke)
Stage 1 -> Stage 2 handoff
Validation MSE (log scale)
4×10^{-3}
3×10^{-3}
2×10^{-3}
10^{-3}
6×10^{-4}
4×10^{-4}
3×10^{-4}
2×10^{-4}
0
20
40
60
80
100
120
140
Epoch